

LenddoEFL: Serving customers better with data-driven algorithms hosted on Google Cloud

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ABOUT LENDDOEFL



LenddoEFL migrated to Google Cloud to serve financial inclusion customers with alternative credit and identity verification services

About LenddoEFL

LenddoEFL's mission is to provide 1 billion people with access to powerful financial products at a lower cost, faster, and more conveniently. The company uses AI and advanced analytics to bring together digital and behavioral data to help lenders confidently serve the underbanked. LenddoEFL has provided credit scoring, verification, and insights to more than 50 financial institutions.

Industries: Financial Services & Insurance

Google Cloud Results

- Improves approval rate, default rate, and processing time
- Enables savings of up to 55% by reducing 160 servers down to 25 servers
- Migrates from the legacy cloud provider to Google Cloud in less than 2 hours

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Location: Singapore

(<https://cloud.google.com/kubernetes-engine/>),

Google Cloud (<https://cloud.google.com/>)

(<https://cloud.google.com/memorystore/>),

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Operations

Cloud Memorystore

(<https://cloud.google.com/memorystore/>)

Monitoring

Cloud Storage (<https://cloud.google.com/storage/>)

Operations

(<https://cloud.google.com/products/operations>)

IAM (<https://cloud.google.com/iam/>)

Logging (<https://cloud.google.com/logging/>)

Monitoring (<https://cloud.google.com/monitoring/>)

Kubernetes (<https://cloud.google.com/kubernetes/>)

Cloud IAM (<https://cloud.google.com/iam/>)

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Millions of consumers around the world cannot access financial services because they have little or no credit history. LenddoEFL

(<https://include1billion.com/>) is the first company to

offer alternative credit scores based on the consumer's digital footprint, from social media posts to geotagged photos, and behavioral data derived from psychometric tests. With the help of analytics and algorithms, LenddoEFL helps financial institutions streamline loan processing and gain access to unbanked consumers.

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—Florentin Lenoir, Chief
of Staff, LenddoEFL

LenddoEFL collects and aggregates traditional data such as borrowing history and non-traditional data such as social media activity and telecom data to gain insight into the consumer. Applicants are also invited to answer 15 to 20 questions in behavioral tests that reveal certain personality traits associated with credit risks. Data from individuals is accessed only with their explicit consent. The company feeds billions of data points into its powerful algorithms to predict a borrower's likelihood to repay or default on a loan.

LenddoEFL delivers real-time results to financial institutions through its client dashboard or API connector. Since its inception in 2011, the company handled 12 million applications from more than 50 clients in 15 countries. In 2019, LenddoEFL announced that it experienced 100 percent year-on-year growth for the past eight years. The company's mission is to provide 1 billion people with access to financial services.

The company relies heavily on its core infrastructure to support exponential growth in the coming years. LenddoEFL migrated from a different cloud to Google Cloud Platform (<https://cloud.google.com/>) for a more reliable and scalable cloud environment that requires little to no maintenance.

"Over the past year or two, our team has been optimizing our infrastructure by relying on Google

Kubernetes Engine. We experienced limitations in terms of services and microservices available in Singapore with our previous provider," says Florentin Lenoir, Chief of Staff at LenddoEFL. "After a careful review of the top cloud service providers in the market, we selected Google due to its expertise in Google Kubernetes Engine, the availability of services in Singapore, and its competitive pricing."

LenddoEFL's platform is currently hosted on Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>) (GKE) using Google Cloud services such as Cloud Storage (<https://cloud.google.com/storage/>), Stackdriver (<https://cloud.google.com/stackdriver/>), Cloud Memorystore (<https://cloud.google.com/memorystore/>), and third-party applications such as MongoDB Atlas, Grafana, and Prometheus.

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According to Florentin, the migration to Google Cloud was smooth, with LenddoEFL handling everything internally. The planning took four weeks from building an instance to testing it on Google Cloud. LenddoEFL stopped overall operations for the scheduled maintenance, and the migration was executed in less than two hours. "Everything worked perfectly," he says. "Our clients didn't feel any impact, and now they're enjoying a more stable platform."

Delivering a smooth application process with Google Cloud

"Before, we had to reserve instances to save costs. By moving to Google Cloud, we're now benefiting from pay-as-you-go pricing and scalability," says Florentin. "We don't always have visibility into which customers' marketing campaigns cause spikes in traffic. Autoscaling means the service remains available at peak demand. We can scale up and down seamlessly, without any IT intervention."

Florentin adds, "Instead of over-provisioning, we pay only for the data we use. In terms of overall efficiencies, we have reduced the number of servers from 160 on our previous cloud platform to 25 on

Google Cloud. That's a key consideration in the vendor selection process."

Previously, the company invested heavily in reserve capacity to handle unexpected spikes. Seasonal activities, marketing campaigns, and new product launches from clients tend to drive traffic to the LenddoEFL platform. Due to such events, the average number of applications a month may increase from 150,000 to 600,000. With its previous cloud provider, LenddoEFL noticed a drop in SLA level during peak traffic.

There is a significant improvement in platform performance since LenddoEFL migrated to Google Cloud. Google Cloud can handle growing volumes of data and traffic fluctuations without compromising user experience. With Google Cloud, memory and CPU resources can be assigned manually or dynamically to minimize downtime.

"Google Cloud
keeps us secure
and compliant
by providing
data encryption
by default and
compliance with
SOC 2 auditing

industry
standards."

—Florentin Lenoir, Chief
of Staff, LenddoEFL

Driving high-value work with Google Kubernetes Engine

"One reason for moving to Google Cloud is Kubernetes (<https://cloud.google.com/kubernetes/>). We want a Kubernetes environment to build and deploy our containerized applications quickly. Google Kubernetes Engine was an obvious choice," Florentin shares. "It eliminates in-house management of Kubernetes clusters so we can focus on value-added work such as improving algorithms and customer experience."

LenddoEFL's IT team used to spend up to 25 hours debugging code and monitoring updates on its previous cloud platform. LenddoEFL now uses Stackdriver Logging (<https://cloud.google.com/logging/>) to view all log data and cluster metrics with a single click, and Stackdriver Monitoring (<https://cloud.google.com/monitoring/>) to quickly identify and resolve bottlenecks in the system and applications.

"Overall, the team found that it was effortless to build stacks using Google Cloud and Kubernetes. Google Kubernetes Engine simplifies the work of the

developers and simplifies the ability to replicate identical stacks efficiently," says Florentin.

Integration with MongoDB Atlas makes it easy for LenddoEFL to deploy machine learning workloads on GKE. The company uses the cloud database to store opt-in social media data from millions of applicants to support credit scoring and identity verification for the financial institutions it works with. Data scientists at LenddoEFL innovate with the company's proprietary algorithms by accessing and analyzing vast amounts of client and external data to deliver new insights for clients.

Protecting data privacy and cybersecurity with Google Cloud solutions

Financial institutions need to adhere to industry regulations, such as FINRA and GDPR, which differ from market to market. The executive team at LenddoEFL works closely with central banks to improve awareness and understanding of data security in the cloud.

"Given that we handle a lot of identity and banking information, it's important that we set a high bar for security to gain trust from consumers and regulators alike," Florentin emphasizes. "Google Cloud keeps us secure and compliant by providing data encryption by default and compliance with SOC 2 auditing industry standards."

LenddoEFL uses Google Cloud security solutions such as [Cloud Identity and Access Management](#)

(<https://cloud.google.com/iam/>) (Cloud IAM) and Role-Based Access Control (RBAC) to authorize user access and protect cloud workloads on GKE. External parties connect to LenddoEFL's virtual environment via a secure VPN to protect against DOS and other risks.

"Because we now manage fewer servers, we have a better view of our architecture. We're able to detect and respond to credible threats more effectively," Florentin and shares.

Harnessing AI-assisted image recognition with Vision API

"Our architecture is now fairly stable. It's time for us to scale up with new features and enhancements. For example, [Vision API](https://cloud.google.com/vision/) (<https://cloud.google.com/vision/>) could help us serve our customers better by quickly processing images with greater accuracy."

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